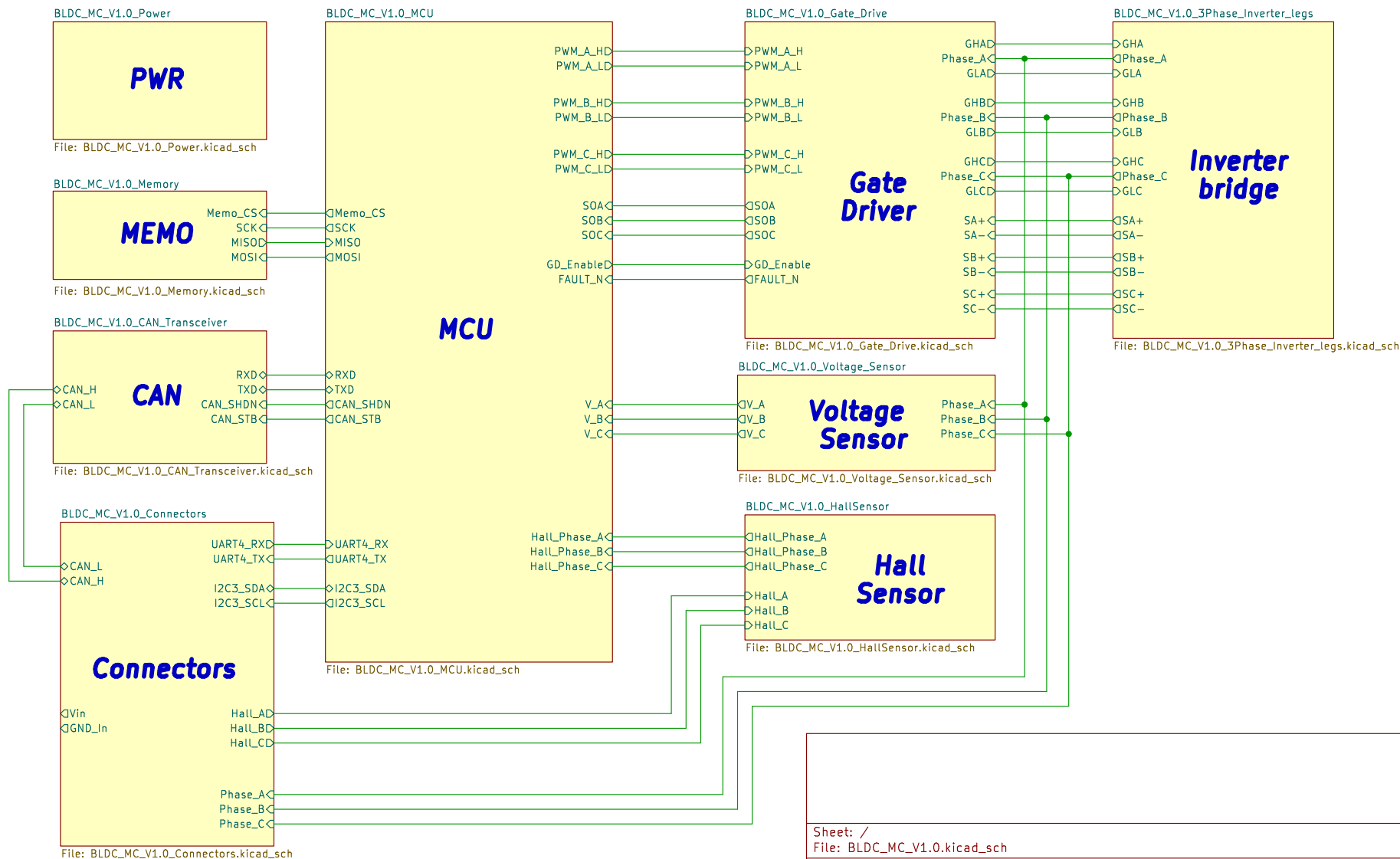


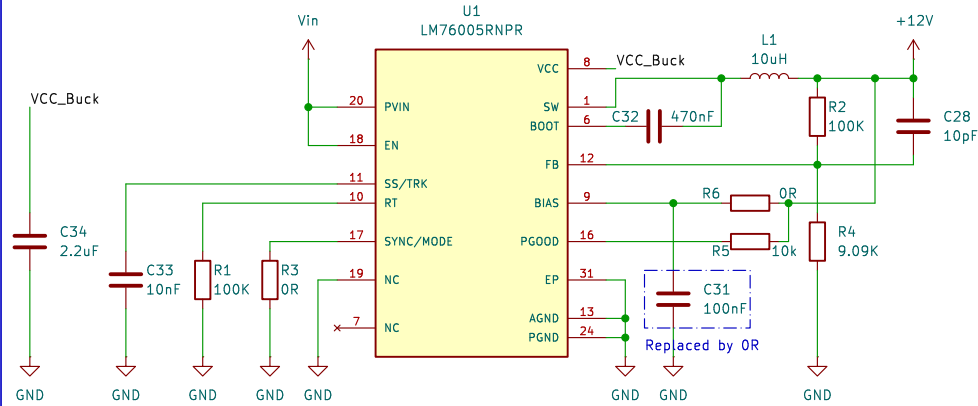
[1] BLDC_MC V1 60V 30A



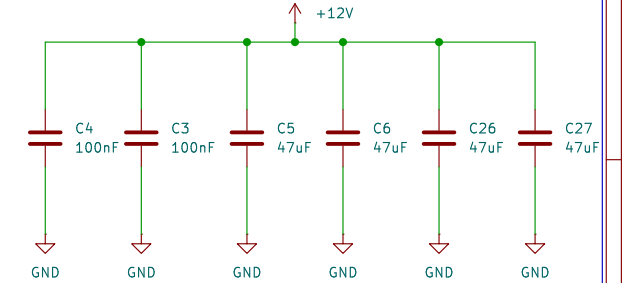
[1] Power Conversion Vin to 12V and 12V to 3.3V

The maximum input voltage of this 60V so we can use for BLDC motors up to 60V

Buck IC Circuit

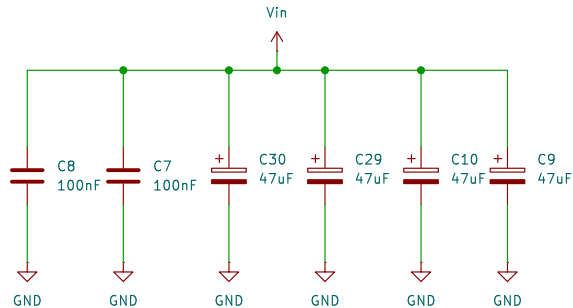


Output Cap Bank



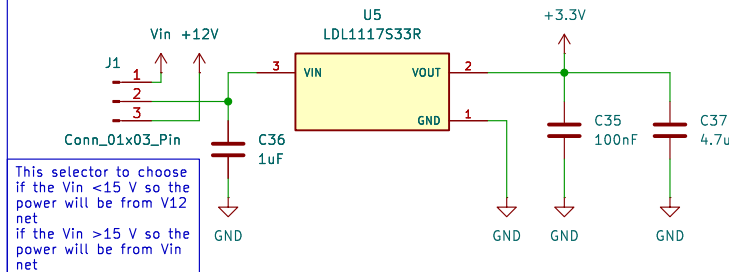
Output Capacitor Bank
Recommended from IC datasheet
Cout = 150uF
Reference design puts 5 of 47uF

Input Cap Bank

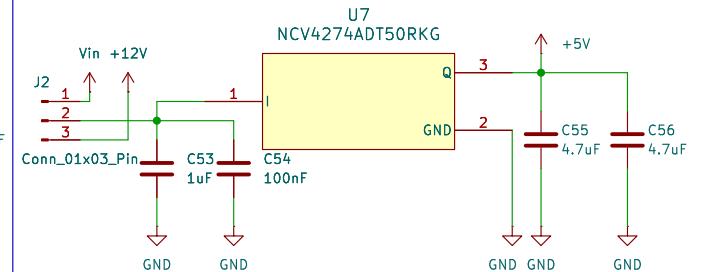


Input Capacitor Bank
To make the input power pure DC

LDO Fixed out 3.3V

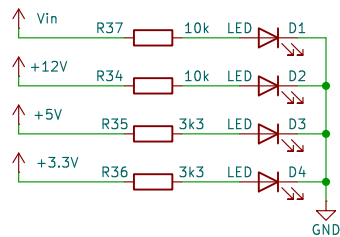


LDO Fixed out 5V



Voltage Level	Usage
12V	Gate drive and MOSFET firing Voltage
5V	for Hall Sensor
3.3V	Controller and aux ICs

Indication Leds



Sheet: /BLDC_MC_V1.0_Power/
File: BLDC_MC_V1.0_Power.kicad_sch

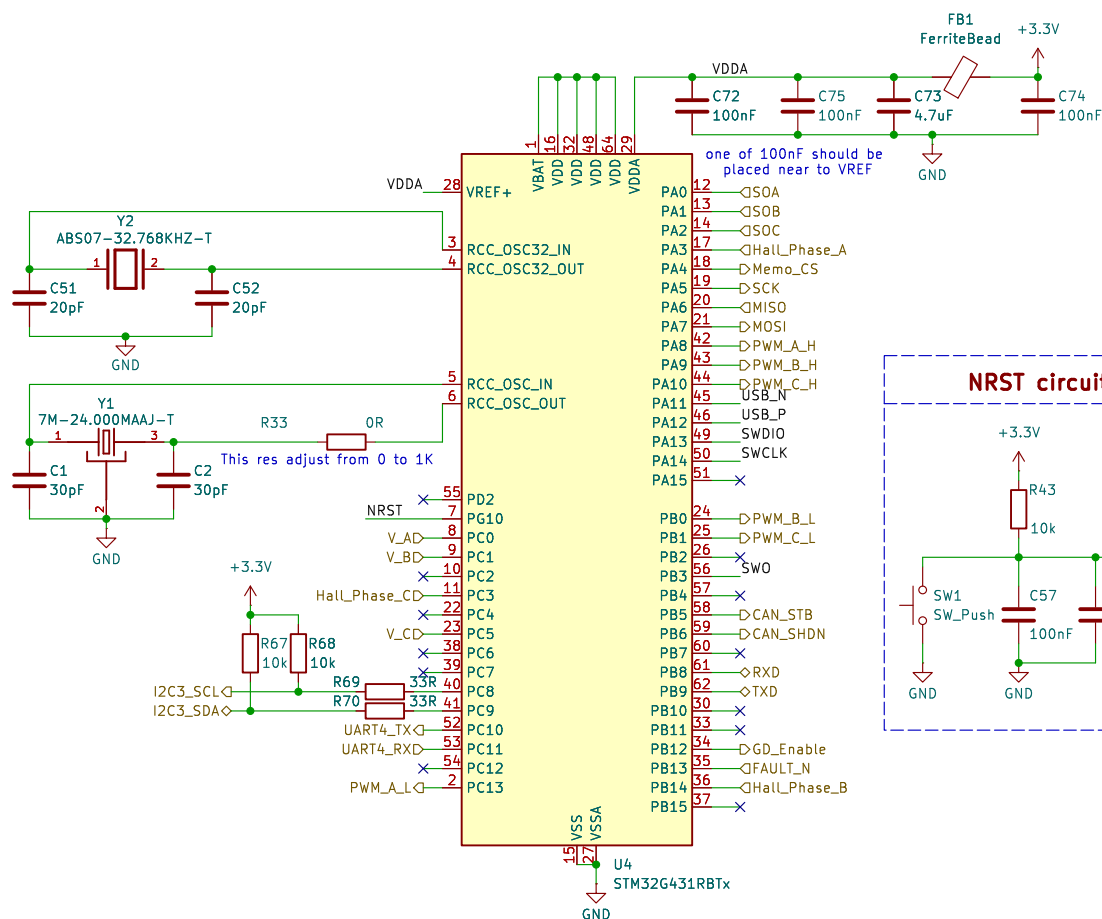
Title: BLDC_MC

Size: A4 Date: 2026-01-05
KiCad E.D.A. 9.0.7

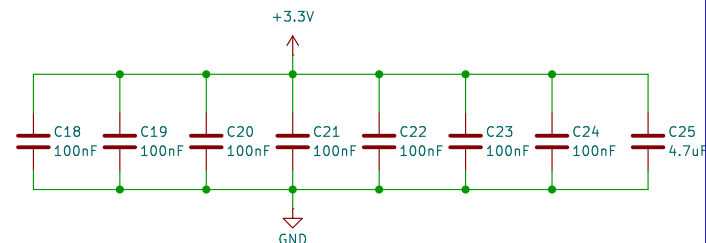
Rev: V1
Id: 2/10

[3] STM32G431RBT6 Controller Interface

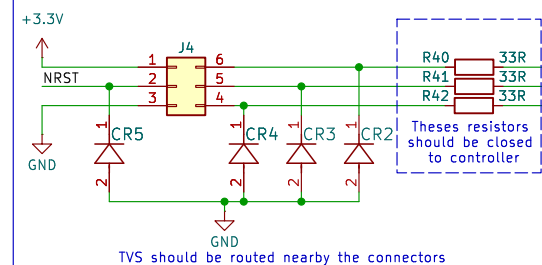
This MCU can handle the FOC control Algorithm



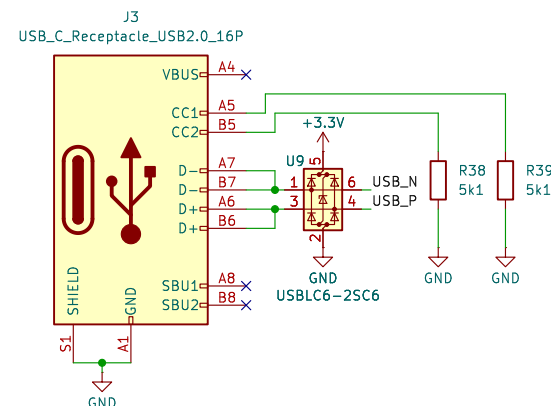
Power bypass Capacitors



Debugger Interface (Serial Wire Debugger)



USB-C



Sheet: /BLDC_MC_V1.0_MCU/
File: BLDC_MC_V1.0_MCU.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.7

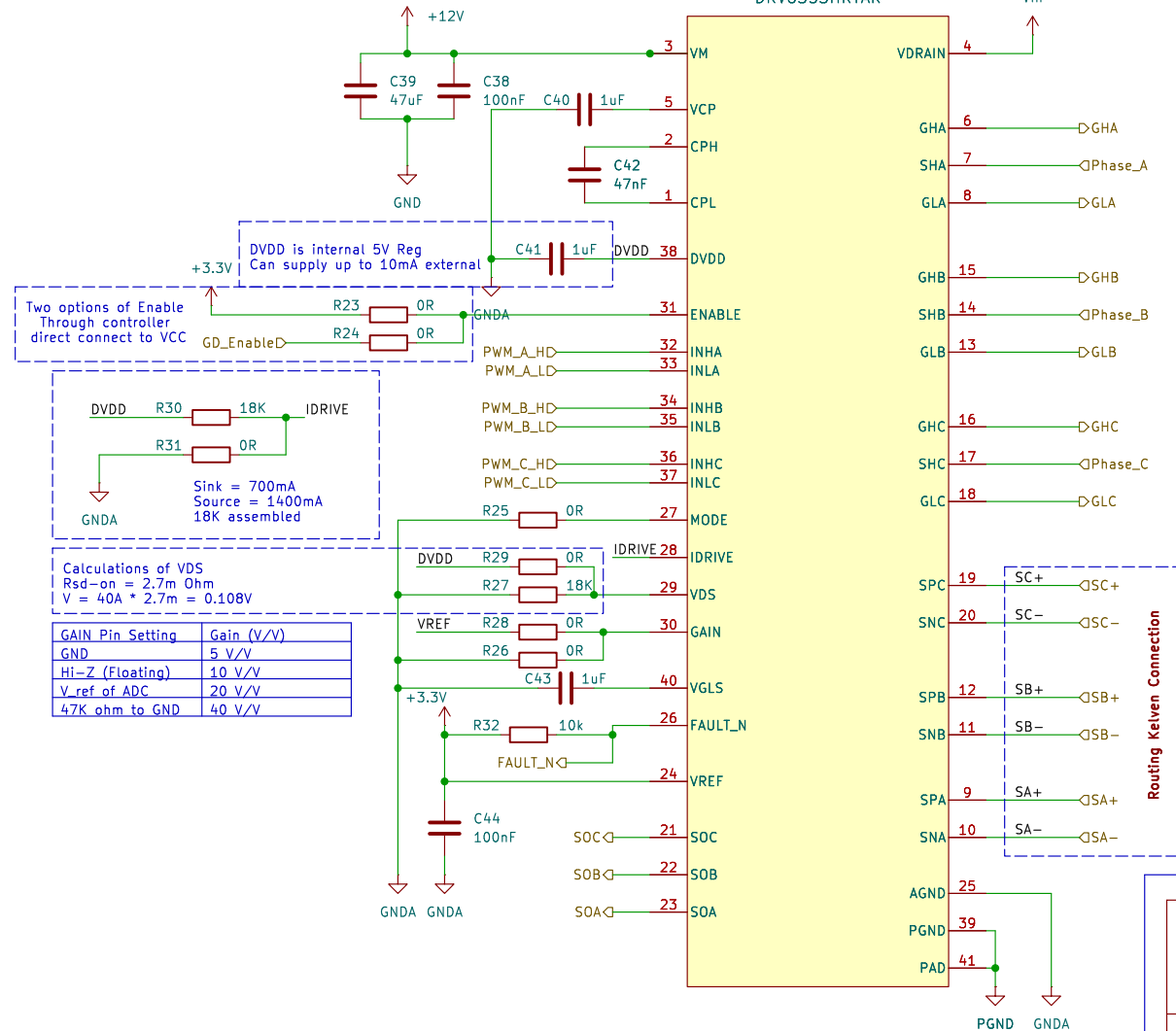
Rev:

Id: 3/10

[4] Smart Gate Drive

for the 3 phase Inverter with current sensor

U6
DRV8353HRTAR



Sheet: /BLDC_MC_V1.0_Gate_Drive/
File: BLDC_MC_V1.0_Gate_Drive.kicad_sch

Title: BLDC_MC

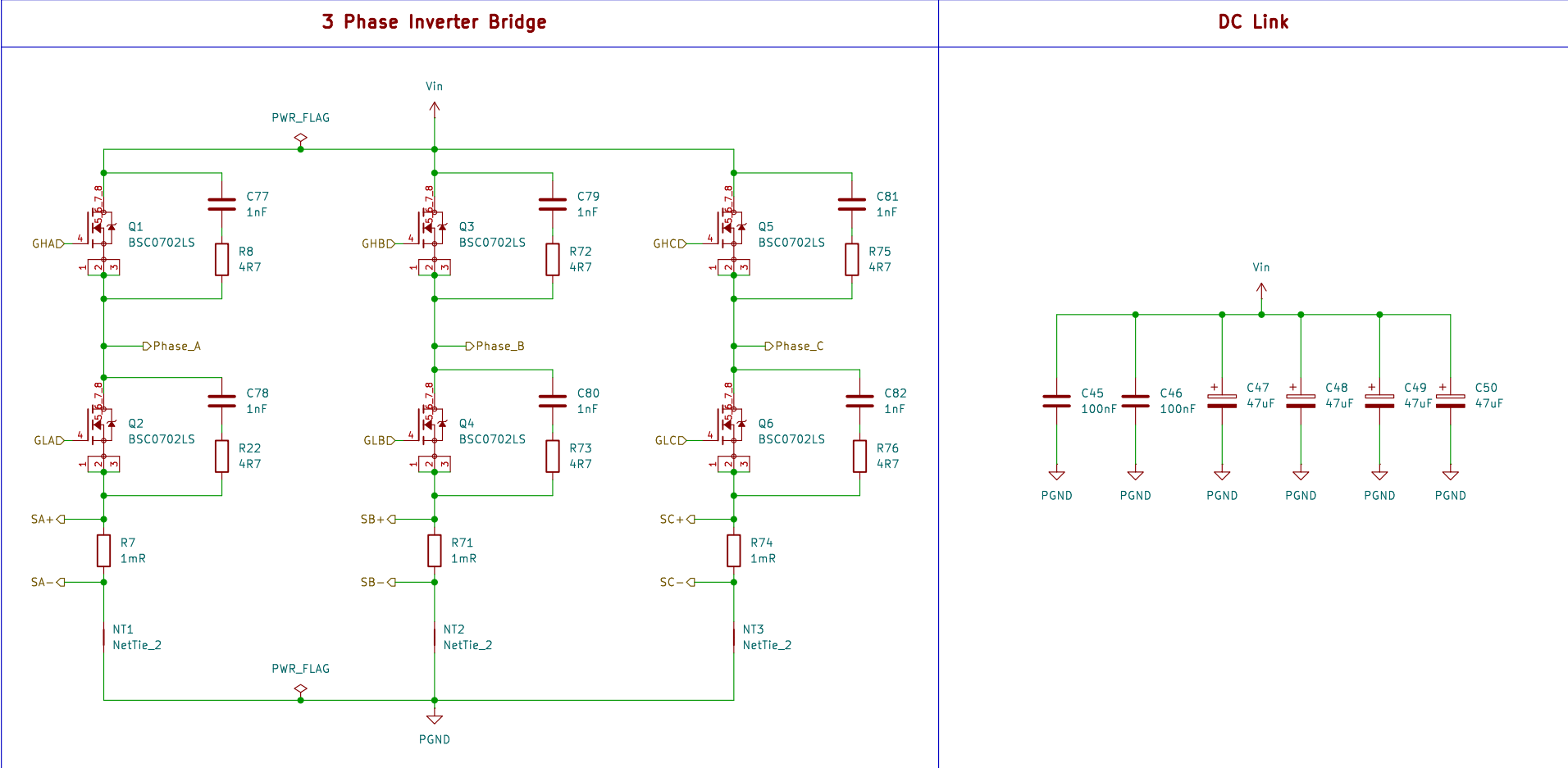
Size: A4
KiCad E.D.A. 9.0.7

Date: 2026-01-12

Rev: V1

Id: 4/10

[5] 3 Phases Inverter Bridge



Sheet: /BLDC_MC_V1.0_3Phase_Inverter_legs/
File: BLDC_MC_V1.0_3Phase_Inverter_legs.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-01-07
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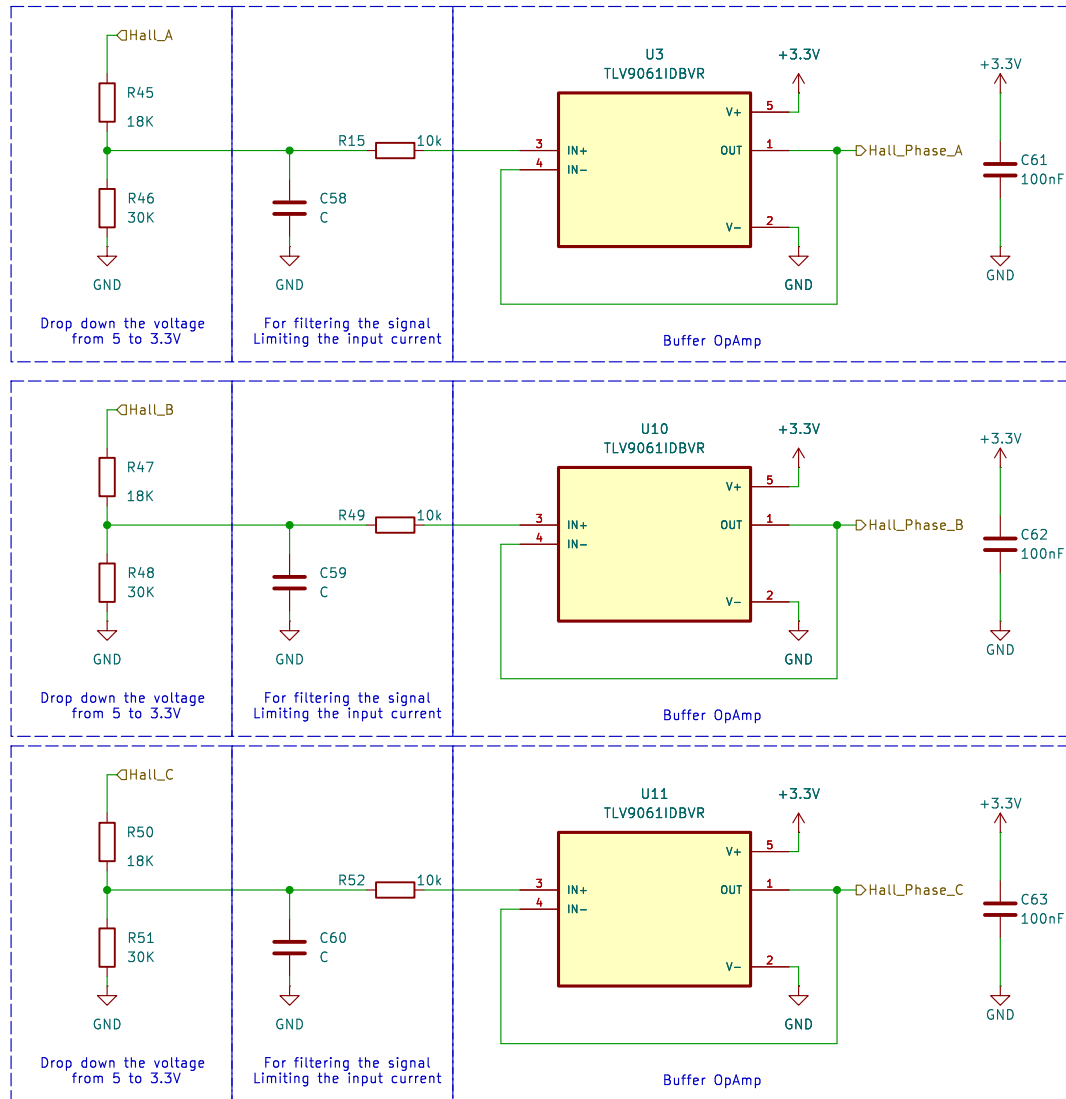
KiCad E.D.A. 9.0.7

Rev: V1

Id: 5/10

[6] Hall Effect Sensor Interface

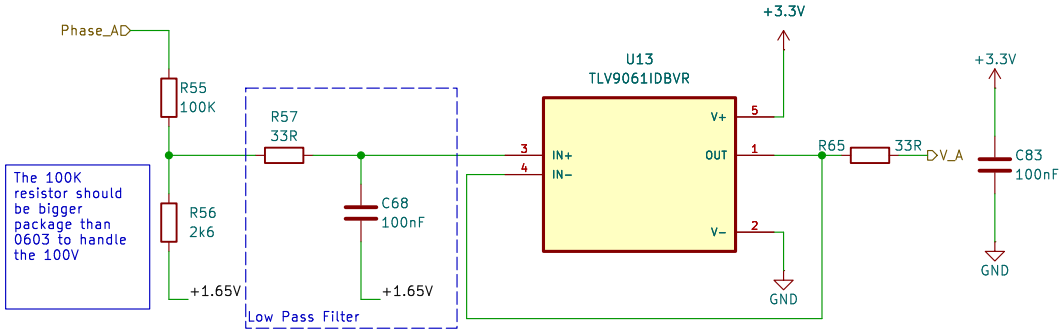
For the 3 Phases of Hall Sensor



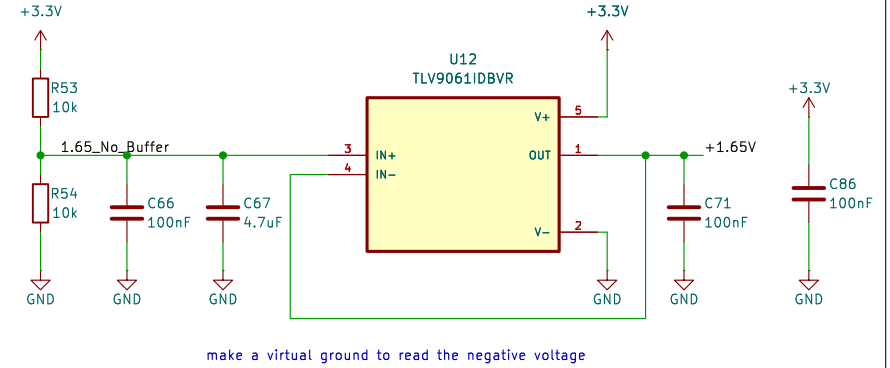
Sheet: /BLDC_MC_V1.0_HallSensor/
File: BLDC_MC_V1.0_HallSensor.kicad_sch
Title: BLDC_MC_V1.0_HallSensor
Size: A4 Date: 2026-03-05
KiCad E.D.A. 9.0.7 Rev: V1
Id: 6/10

[7] 3 Phase Inverter Voltage Sensing

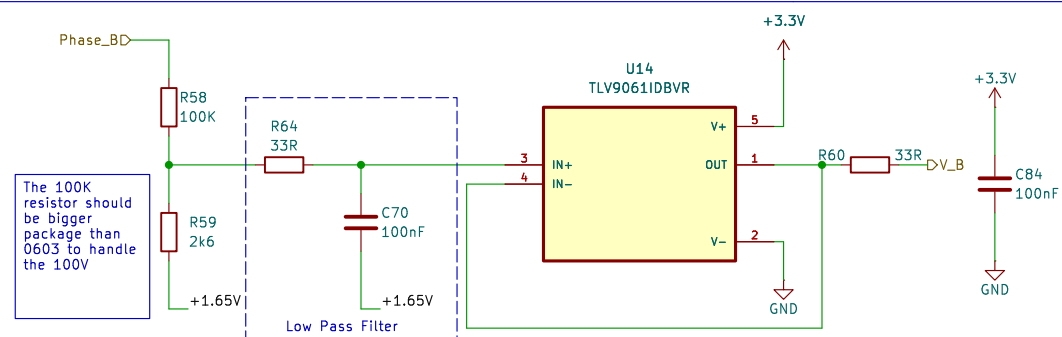
Phase A



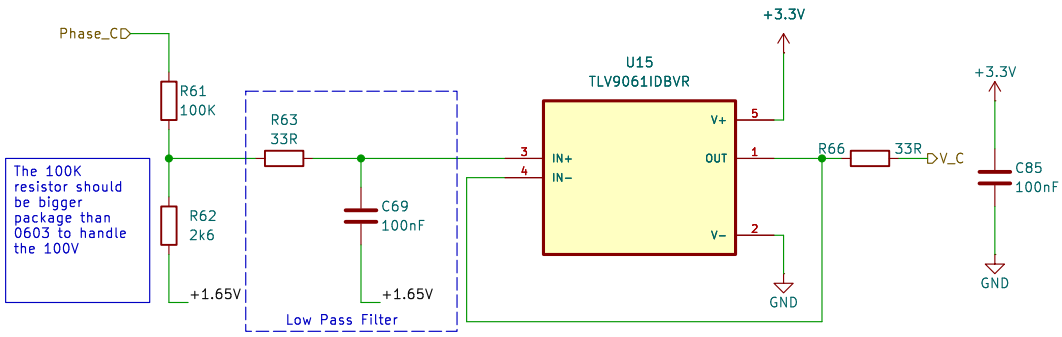
Virtual Ground +1.65V



Phase B



Phase C



Sheet: /BLDC_MC_V1.0_Voltage_Sensor/
File: BLDC_MC_V1.0_Voltage_Sensor.kicad_sch

Title: BLDC_MC_V1.0_Voltage_Sensor

Size: A4 Date: 2026-03-11

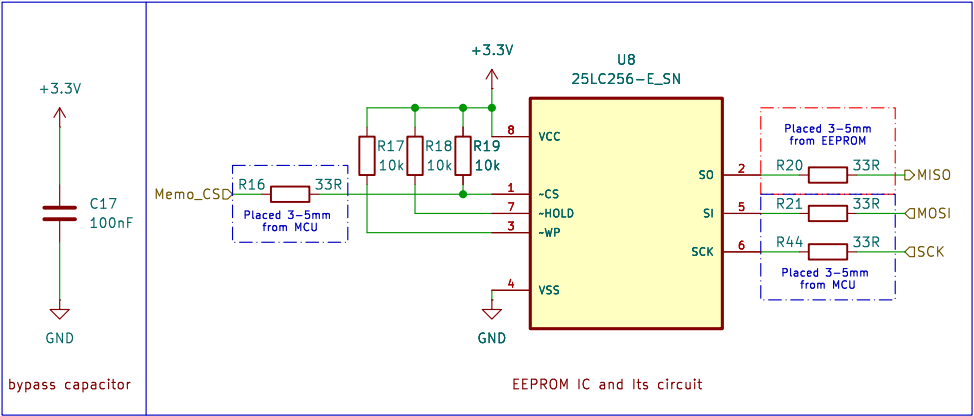
KiCad E.D.A. 9.0.7

Rev: V1

Id: 7/10

[8] EEPROM Memory 256Kbit

This memory used to store important data and configration parameters



Sheet: /BLDC_MC_V1.0_Memory/
File: BLDC_MC_V1.0_Memory.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-01-07

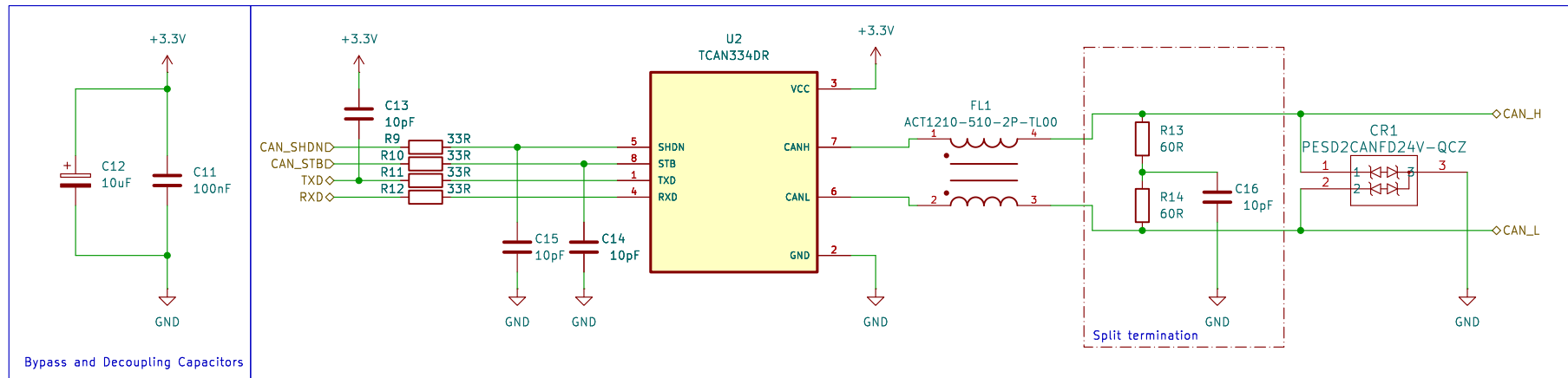
KiCad E.D.A. 9.0.7

Rev: V1

Id: 8/10

[9] CAN Transiver Classic CAN 1Mb/S

This IC can operate as CAN-FD for faster data rate



Note

Series Resistors Help manage signal integrity by damping potential high-frequency noise or reflections on the digital traces.

Parallel Capacitors is used to filter noise on the digital I/O lines

Sheet: /BLDC_MC_V1.0_CAN_Transceiver/
File: BLDC_MC_V1.0_CAN_Transceiver.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-01-05

KiCad E.D.A. 9.0.7

Rev: V1

Id: 9/10

[10] Board's Connectors Power & Communication

